

BME 1002 "Introduction to Biomedical Engineering"

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Office hours: Tue/Thur 9:30 AM or
by appointment
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Class hours: 3:20-5:35 Tue in Engineering Building E200

Course Content

BME 1002 provides an overview of biomedical engineering, principles of living systems, engineering applications of bio-molecular and cellular fundamentals, and medical engineering. The scope of biomedical engineering and the biomedical engineer's role in society will be discussed. Weekly seminars will be organized as a lecture series. Biomedical Engineers from industry and academia will lecture on topics such as sub disciplines of biomedical engineering, biomedical ethics, regulations, biomedical advancements, job opportunities and other timely biomedical activities.

Course instructional format

The class will have lectures, class activities, group discussion, and invited speakers. Students will be assessed based on binders, project reports, presentations and in class and group participation. This is a two credit class consisting of one 150 minute class per week.

Recommended Textbook

Biomedical Engineering: Bridging Medicine and Technology Edition 2, W. Mark Saltzman.
Cambridge University Press, 2015. ISBN: 9781107037199

Learning objectives

- Describe the field of biomedical engineering and list the subdisciplines that are included in and their various areas of focus.
- Distinguish between the study of biomedical engineering and the study of human physiology.
- Identify changes in medicine that have enhanced life span and quality of life.
- List examples of engineering in defining or advancing medical treatments.
- Summarize important issues related to a real world biomedical engineering ethical dilemma.
- Demonstrate the steps in the engineering process and describe the purpose of each.
- Produce basic assembly drawings and written assembly instructions for a design project.
- Identify the use and application of important engineering tools, including computers, CAD, word processing and spreadsheet programs.
- Give examples of some of the applications of x-ray, ultrasound, nuclear medicine, optical, and magnetic resonance imaging (MRI) modalities and biosensors.
- Explain how tissue engineering has emerged as a possible solution for organ replacement or healing.
- Distinguish the various types of biomaterials that are available and their common uses.
- Outline the range of medical devices; from artificial hearts and valves to drug-eluting stents that are now used to treat heart disease.
- Describe the relationship between structure, function and material properties in the mechanical behavior of biological tissues.
- Explain the value of the new approaches, based on our understanding of the molecular and cellular biology of cancer, for creating biological treatments.
- Produce effective written and verbal communication in the form of a team report and presentation.

Grading

- **15% Attendance and Class Participation**
 - It is important that students attend class. Missing days will hurt your grade in this course. Please contact the instructor for any and all anticipated absences, and especially emergencies relating to class attendance. Participation in class discussions, asking and answering questions, and helping other class members during group projects is also a critical part of this course.
- **10% Binders or Digital Folder**
 - Organization skills are important for all practicing engineers. Successful engineers keep their work well organized. Binders are a simple way to structure organization into your work. Digital files could also be organized on a computer folder. All student binders/digital folders will be reviewed during the semester and at the end of the term for a grade. Keep all course materials organized and up-to-date. **Save all handouts, assignments and references that you acquire during the semester!**
- **25% Seminar Discussion Notes**
 - Each student will submit a one-page summary of the weekly seminar held the previous week. Format of the notes is posted on Canvas.
- **10% Presentations**
 - Each student or group will present selected topics and answer questions from the instructor and other students.
- **40% Project Reports/ Assignments**
 - Each student or group will produce written reports documenting the procedures followed and final outcome in design and research projects and other homework assignments.

Grading Scale

A	≥ 93%
A-	90-92.9%
B+	87-89.9%
B	83-86.9%
B-	80-82.9%
C+	77-79.9%
C	73-76.9%
C-	70-72.9%
D+	67-69.9%
D	60-66.9%
F	<60%

Attendance Rubric

Participation Grade for Semester	Perfect Attendance	1 Absence	2 Absences	3 Absences	4 Absences	5+ Absences
A	15%	15%	13%	11%	9%	Fail
B	13%	13%	11%	9%	7%	Fail
C	11%	11%	9%	7%	5%	Fail
D	9%	9%	7%	5%	3%	Fail
F	7%	7%	5%	3%	0%	Fail

A = Fully engaged in class and team discussions. Asked and answered questions from instructor and team

members.

B = Mostly engaged in class and team discussions. Occasionally asked and answered questions from instructor and team members. Occasionally distracted or engaged in other activities.

C = Only partially engaged in class and team discussions. Only responded when directly asked a question by the instructor. Occasionally distracted or engaged in other activities.

D = Rarely participated in class and team discussions. Only partially able to answer questions directly posed by professor. Frequently distracted or engaged in other activities.

F = Attended class but did not actively participate. Unable to answer questions directly posed by professor.

For program assessment purposes, anonymous copies of submitted student work may be kept on file.

Course conduct

Students should read, understand and comply with the Lawrence Technological University Student Code of Conduct. For more information see:

http://www.ltu.edu/student_affairs/student_conduct.asp

LTU requires every individual must wear a face covering over their nose and mouth when in any indoor public space. Public space is any shared or communal space such as a classroom, but not limited to, hallways, conference rooms, atriums, and laboratories.

Students who choose not to wear a face covering (mask) are in violation of the LTU Student Code of Conduct, Section C, Rules and Regulations 9. Violators will be sanctioned. Multiple violations will result in loss of privileges and suspension.

Academic Integrity

Violations of academic ethics will be handled according to University policies as set forth in the University's Academic Honor Code see:

http://www.ltu.edu/currentstudents/honor_code.asp

Specific policies for assignments in this course are:

- All undergraduate students are required to write the following pledge on all academic work submitted along with a signature. If the pledge does not accompany an assignment, the faculty can choose to not grade it.
"I have neither given nor received unauthorized aid in completing this work, nor have I presented someone else's work as my own."
- Students may discuss assignments with others, but should complete their own work and write up their own solutions. All assignments and reports must be by individual or group effort only. Plagiarism, without appropriate reference citation, will not be tolerated.

Note: Unforeseen circumstances may necessitate changes, additions, or deletions to this syllabus during the term.

Semester schedule (Subject to change)

Week	Date	Topic (Online Video)	In Class Activities/ Assignment	Speaker (Zoom Livestream)
1	1/19/21	Introduction to BME	Ice Breaker, HW 1	Eric Meyer, Dental Implants
2	1/26/21	Keys to Success	Goals Worksheet	
3	1/2/21	Design (Connections)	Medical Technology, Midterm paper	
4	2/9/21	Innovation (Creativity)	Design Thinking	
5	2/16/21	Sustainability (Creating Value)	Activities of Daily Living	
6	2/23/21	Biomolecules	HW 2	
7	3/2/21	Biomaterials	HW 3	
8	3/9/21	Tissue Engineering	HW 4	
9	3/16/21	Maker Skills	Group Project	
10	3/23/21	Bioinstrumentation	Group Project	
11	3/30/21	Medical Imaging	HW 6	
12	4/6/21	Biomechanics	Final Presentation	
13	4/13/21	Biomedical Modeling	HW 7	
14	4/20/21	Prosthetics		
15	4/27/21	Final presentations	Notebooks Due	

Additional required participation events: (Choose one to attend and turn in seminar summary form)

March, 2021 **Cisler Lecture** 7:00 pm Science Auditorium topic TBD

TBD **LTU Research Day** the poster session is at 12:30-3:00 pm in A200. See www.ltu.edu/provosts_office/research_day for more details.

TBD **LTU Life Sciences Advisory Board Poster Session** see <http://ltu.edu/blogs/biomedstudents/> to view the BME Senior Projects and choose one to summarize.

May, 2021 **BME Senior Projects Final Presentations**